University of Colorado **Boulder**

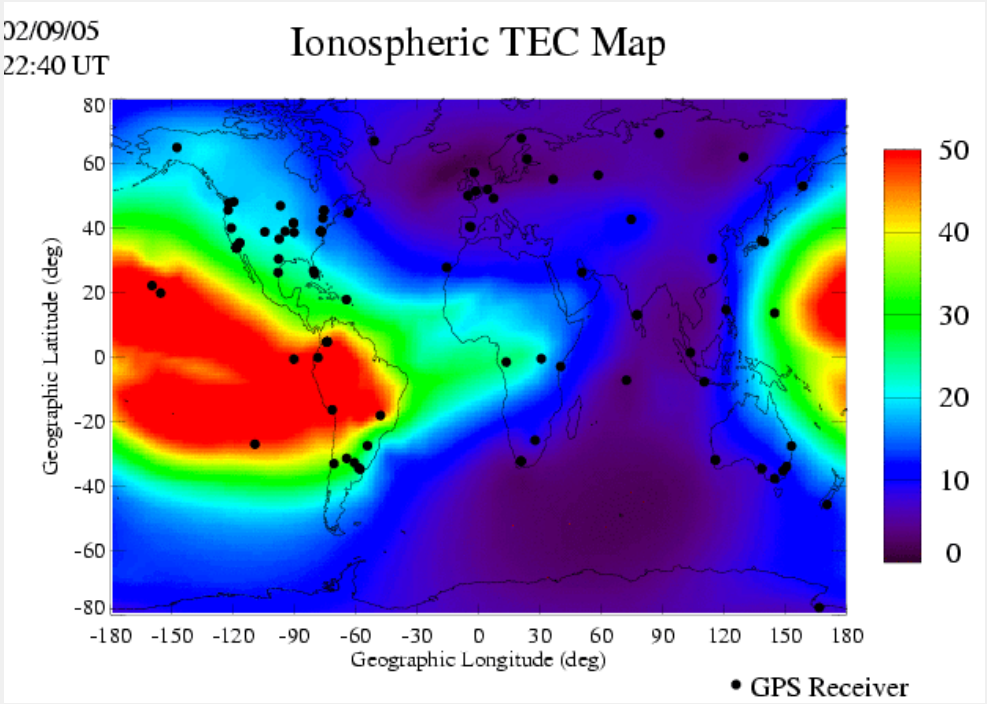
The Ionosphere: Missions and Movies

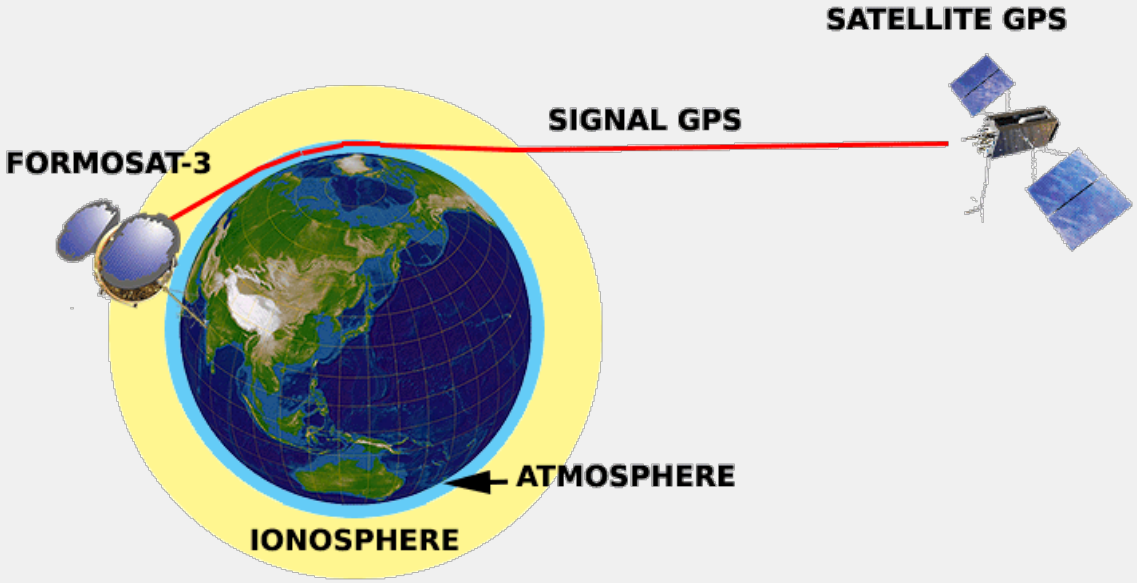
Ionosphere Missions

- **ICON** (Ionospheric Connection Explorer)
  - Using airglow to infer ion densities and winds
- **Ampere** (Active Magnetosphere and Planetary Electrodynamics Response Experiment)
  - Using magnetic fields to measure currents
- **COSMIC** (Constellation Observing System for Meteorology, Ionosphere, and Climate) and **GPS**
  - Measuring the ionosphere using radio occultation / path delay
- *Hidden Figures* and *Apollo 13*: Re-entry comm problem

GPS and COSMIC

- Use radio occultation (path delay) to measure TEC
  - Change in '**path length**': $\Delta l_{\text{iono}} = -\frac{40.3}{f^2} TEC$
- Take a ton of TEC measurements and throw them into a giant inversion model to get 4D ionosphere
- Also use spacecraft-to-spacecraft measurements: COSMIC





COSMIC is GPS receiver instrument on Taiwanese FORMOSAT-3 spacecraft
Takes 1500 radio occultation measurements per day (how is that possible, 1500?)

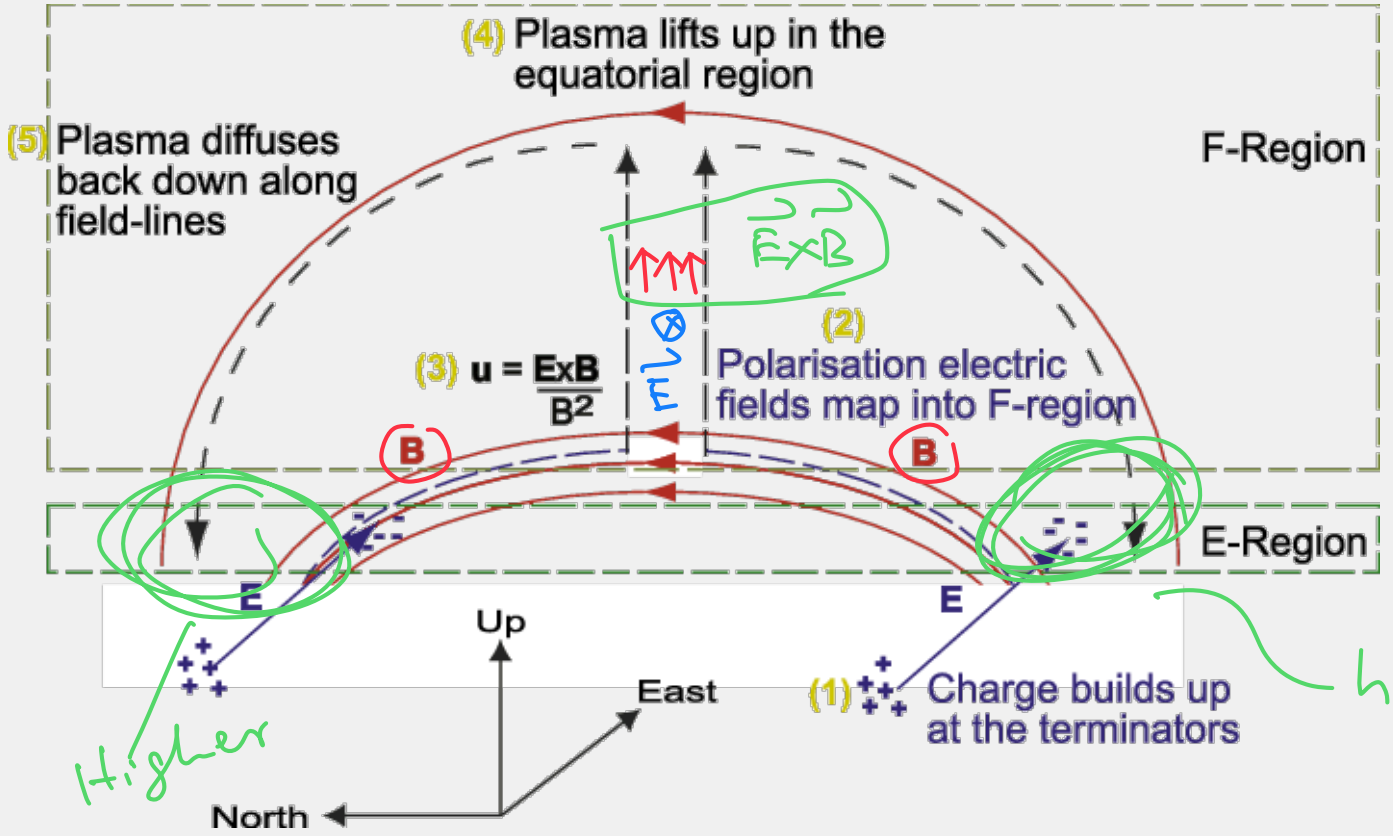
Equatorial (ionization) **Anomaly**

- Due to the equatorial electrojet (EEJ) current, an electric field arises
- E x B leads to a **drift** (known as E x B drift) in the vertical direction
- Plasma rises, but then above some altitude, falls back down along field lines
- "Fountain effect"

$$\vec{v}_d = \frac{\vec{F} \times \vec{B}}{qB^2}$$

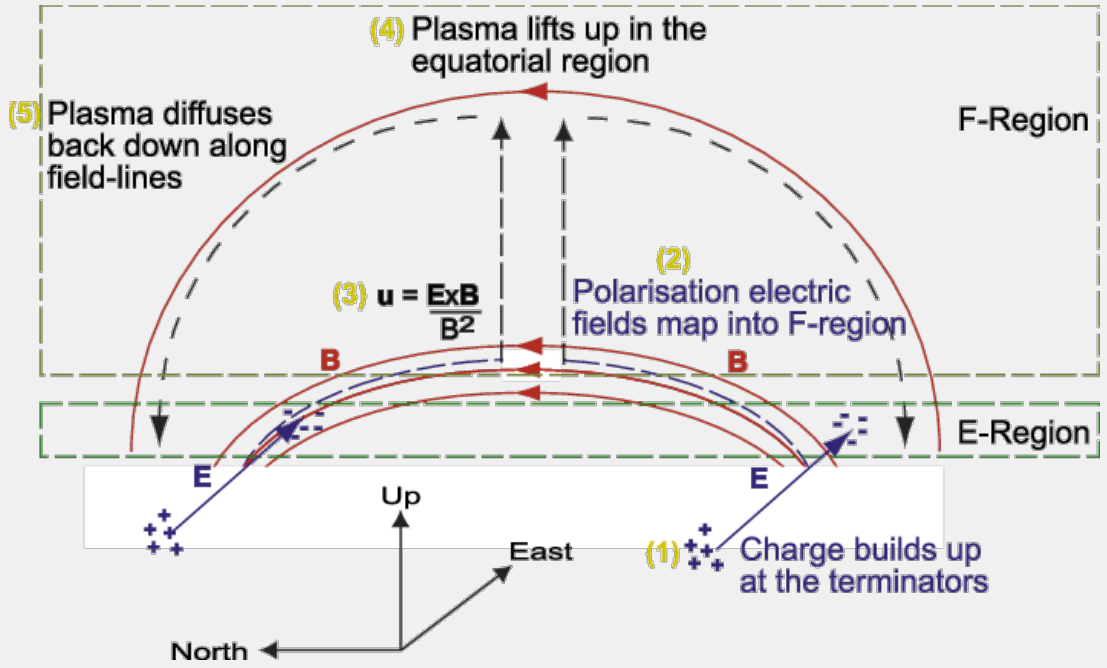
$$\vec{F} = q\vec{E}$$

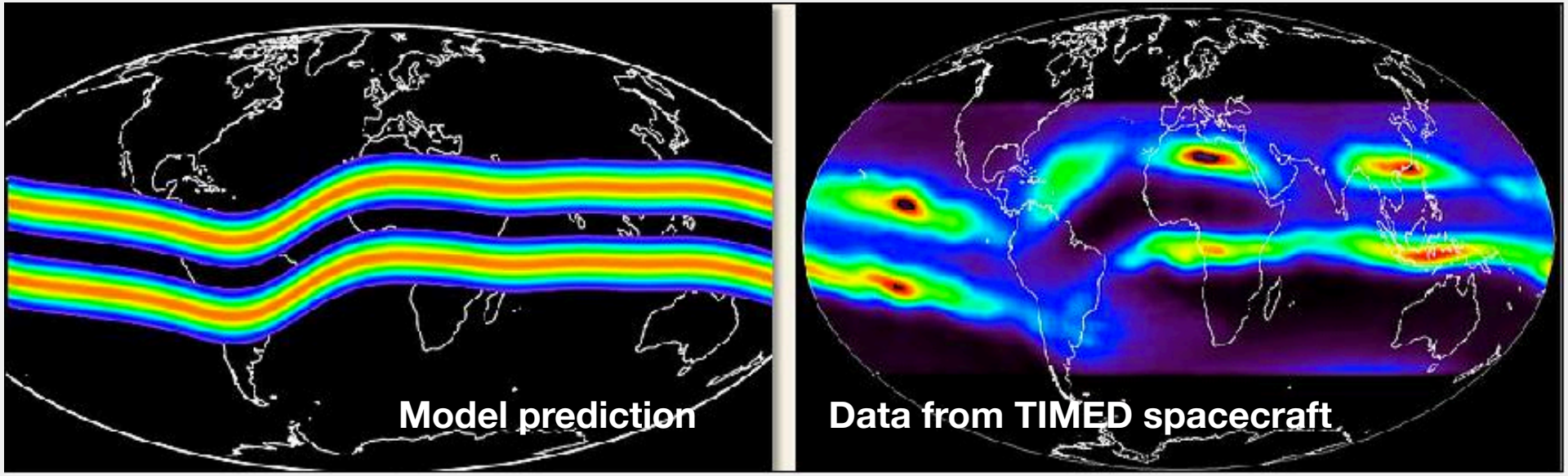
$$\vec{v}_d = \frac{\vec{E} \times \vec{B}}{B^2}$$



The Equatorial Ionosphere

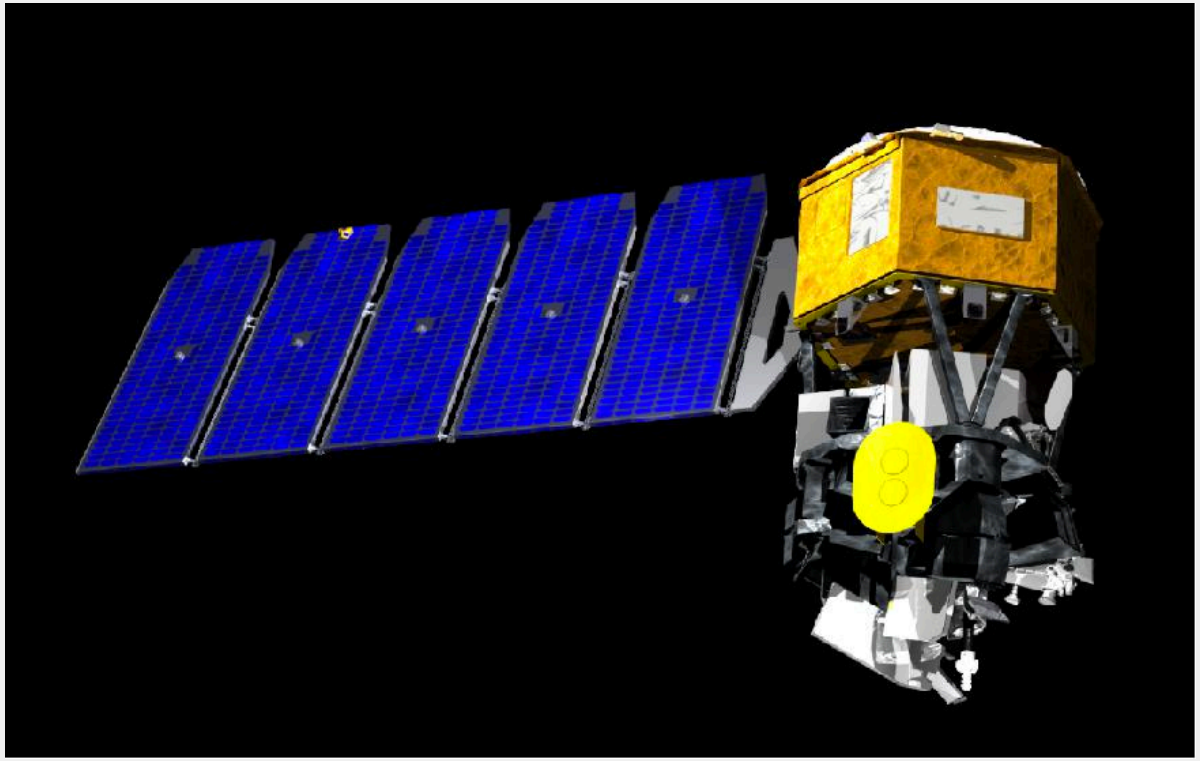
- The Fountain effect predicts two bands of enhanced electron density above and below the magnetic equator
- The truth? A bit more complicated….





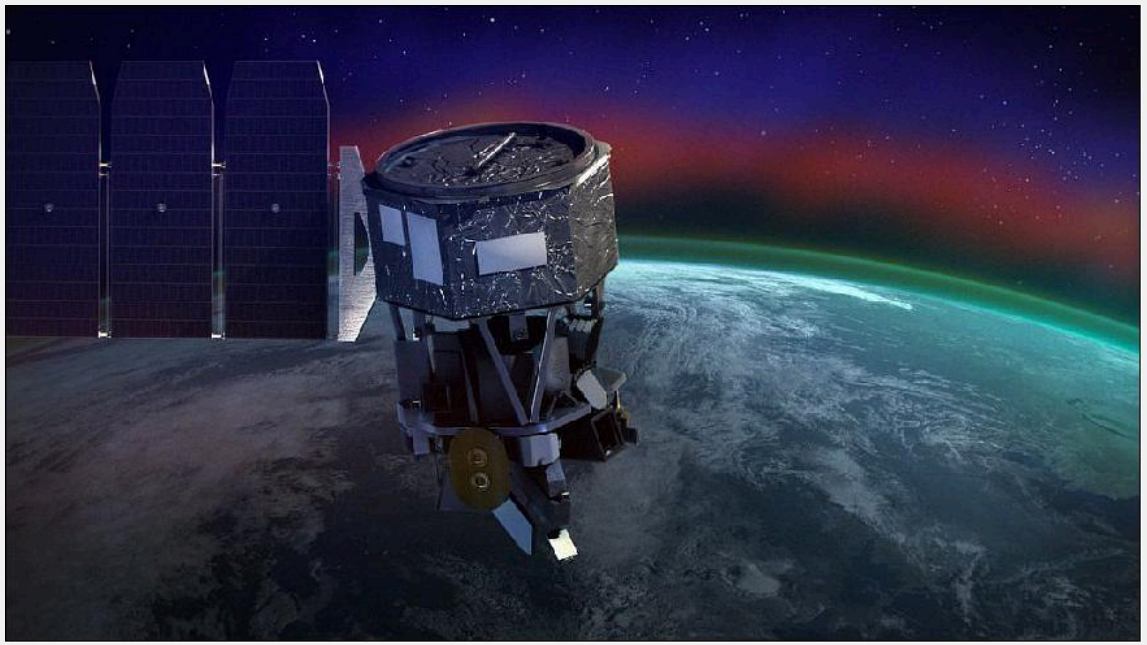
ICON Science and Spacecraft

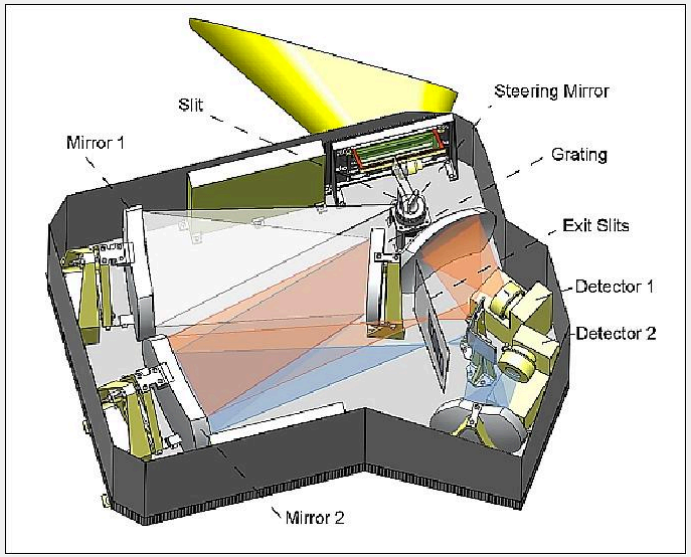
- Understand drivers of ionospheric variability
- Explain how energy / momentum from lower atmosphere reach the space environment (e.g., gravity waves!)
- Explain how drivers create extreme conditions observed during solar-driven geomagnetic storms

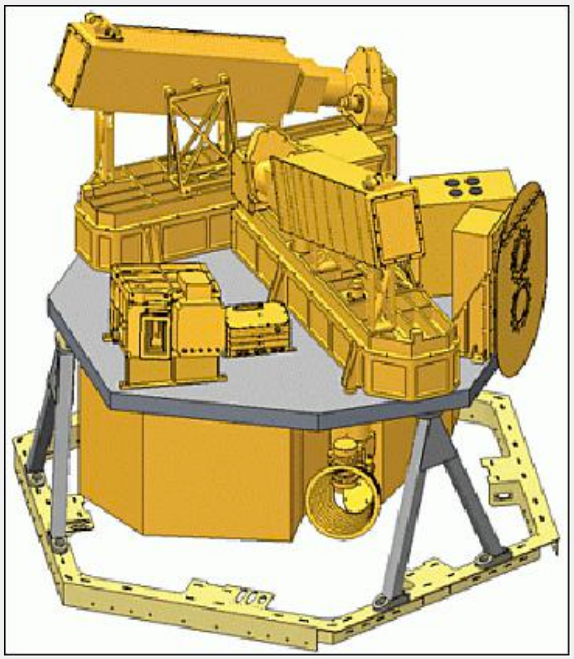


ICON instrumentation

- Four main instruments:
  - **MIGHTI** is a Michelson Interferometer to measure winds and temperatures in the thermal sphere
  - **FUV** is an FUV (135.6 and157 nm) imager; observes UV emissions of N_2 and O to determine O/N_2 ratio
  - **EUV** images 83.4 nm emission from O; resonantly scattered by O+: gives ion density
  - **IVM** is the ion velocity meter; uses a Retarding Potential Analyzer (RPA) to measure relative velocity of ions, therefore winds, as well as temperature and density





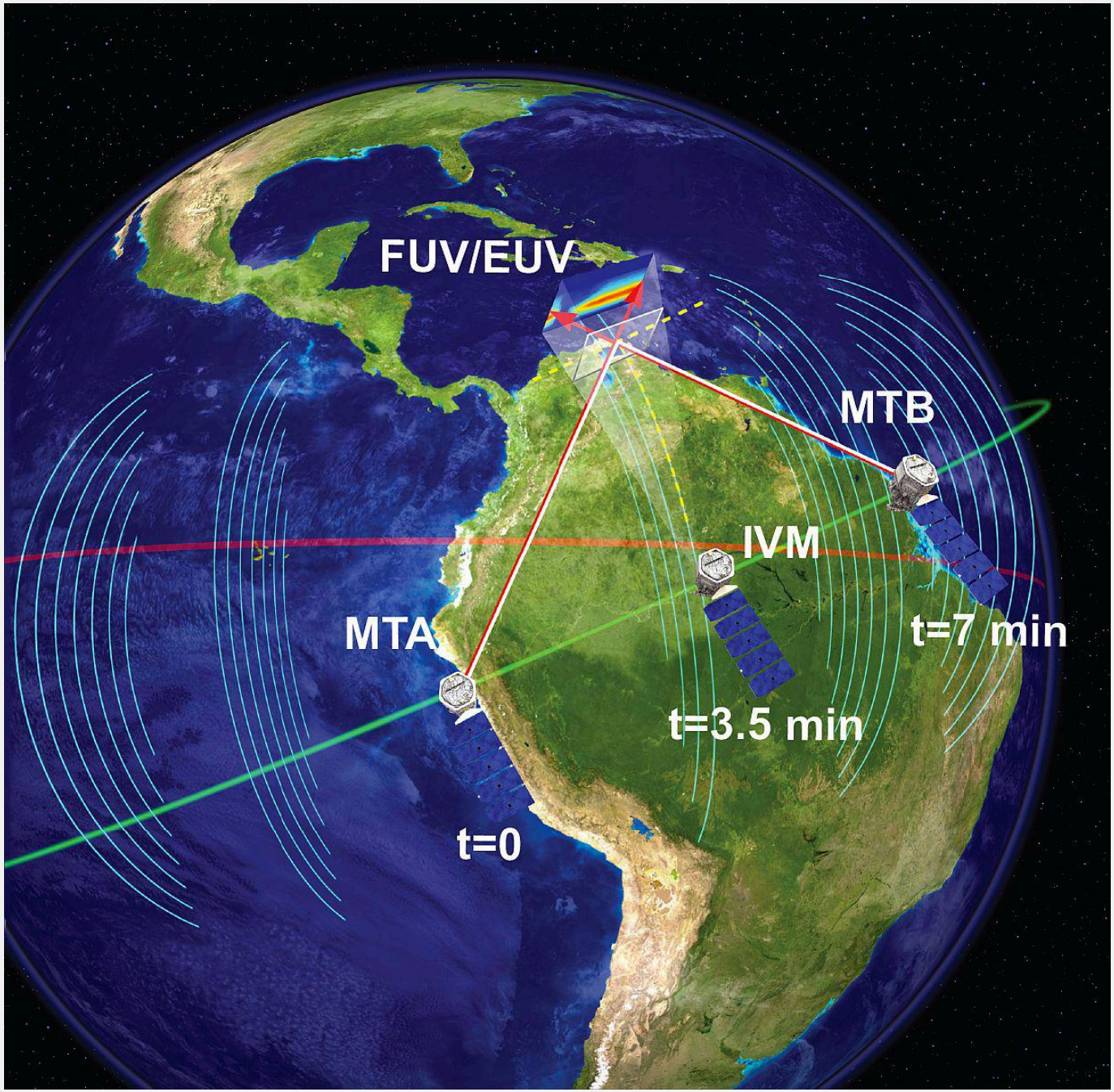


How do you get wind & temperature from optical emissions?

- Line emissions: atomic oxygen has emission lines at 630.0 nm and 557.7 nm (red and green lines)
- If the O atoms were completely stationary, emissions would always be at exactly these wavelengths
- O atoms are not stationary; have some velocity in random directions due to the gas temperature
- Atom moving at v', relative to observer, will emit a photon that is Doppler shifted in wavelength by

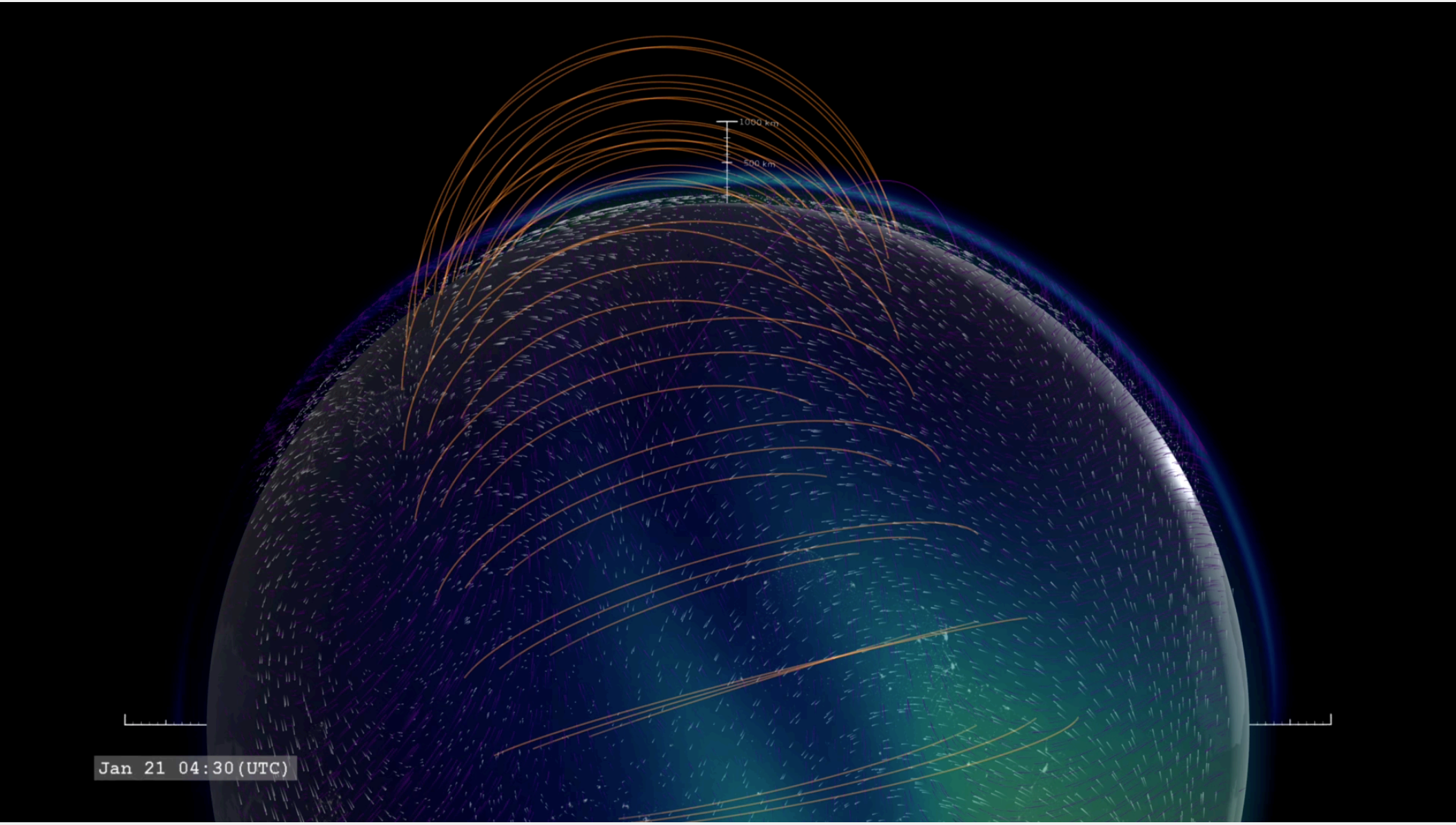
- Add up all the different v's in the Maxwell-Boltzmann distribution, and you get a distribution of wavelengths that is Gaussian
- Line broadening
- Line gets broader when **temperature** is higher
- Need a really good spectrometer to resolve this line!

ICON CONOPS



ICON: 27 deg inclination. MTA: MIGHTI-A, MTB: MIGHTI-B

ICON CONOPS made awesome



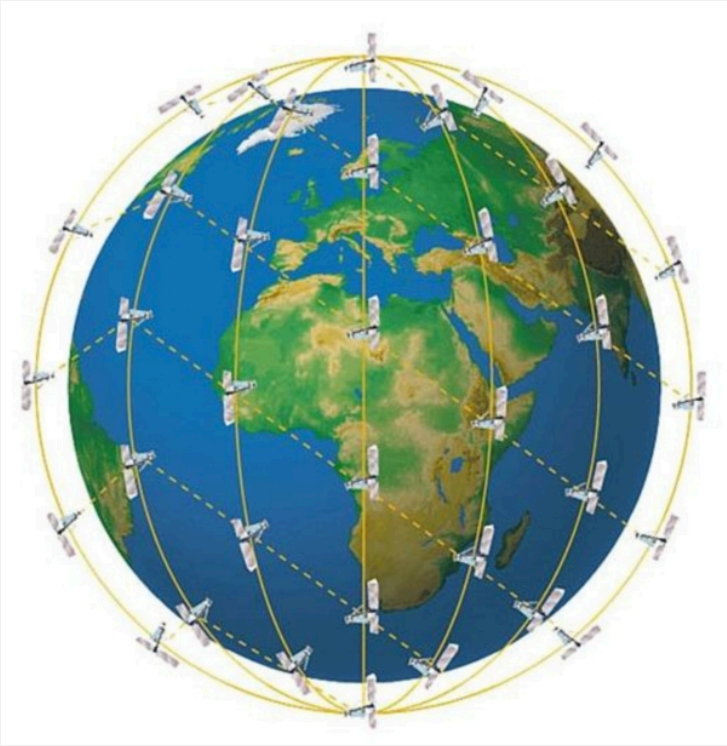
A word from our sponsor (NASA)

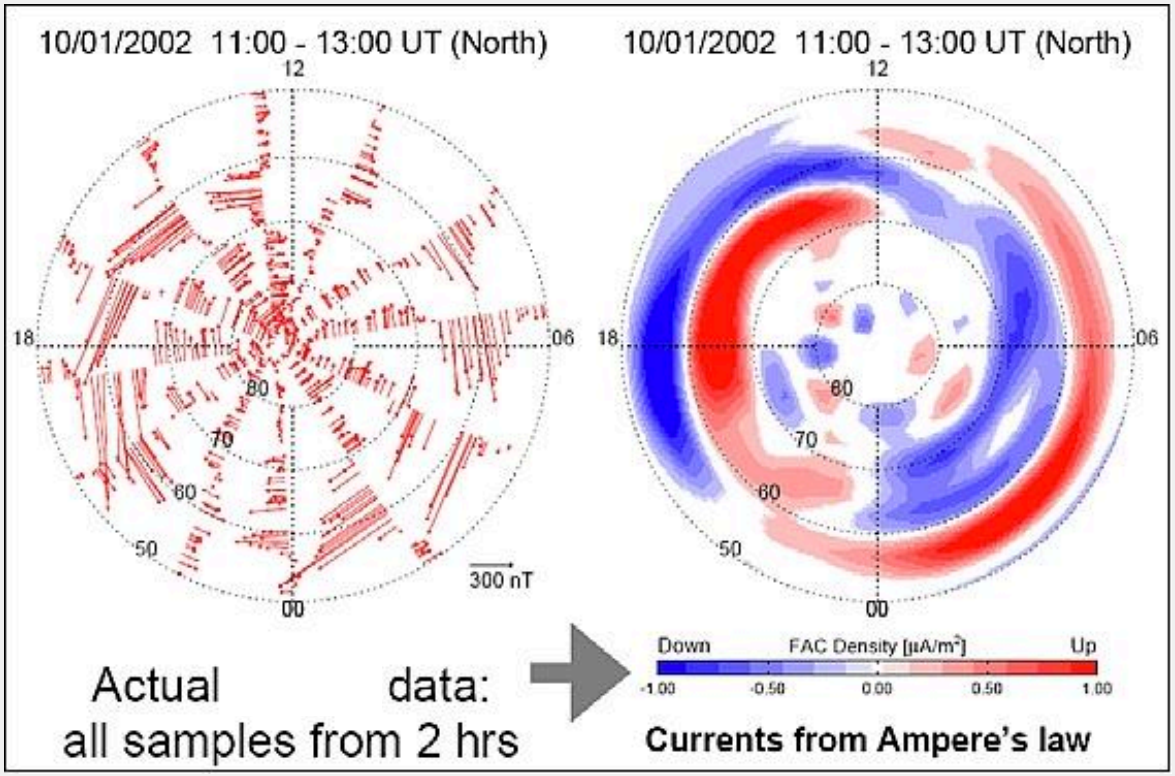
AMPERE

- Recall Ampere's Law: says that currents produce magnetic fields
- AMPERE uses measurements of magnetic fields to determine the currents that created them

$$\mathbf{J} = \nabla \times \mathbf{H}$$

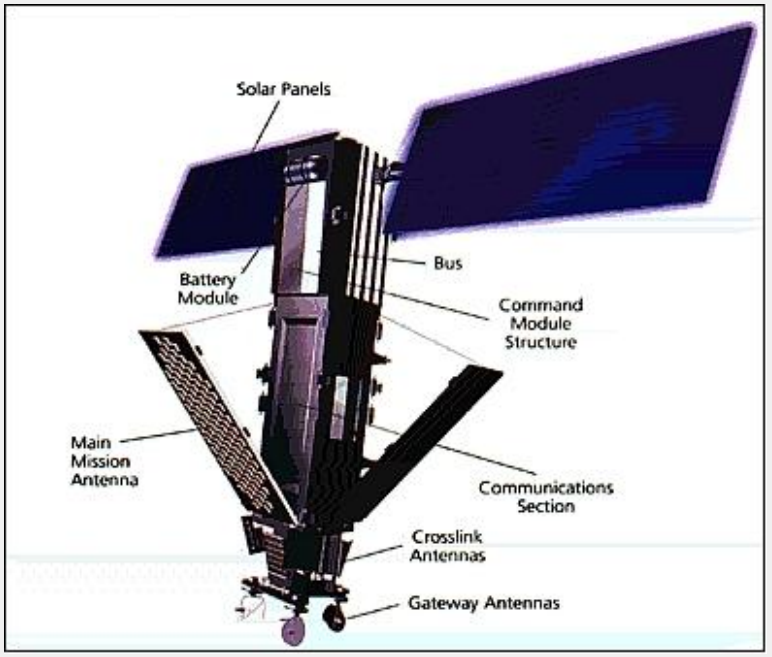
$$\int_C \mathbf{H} \cdot d\mathbf{l} = \int_S \mathbf{J} \cdot d\mathbf{A} = I_{\text{enc}}$$

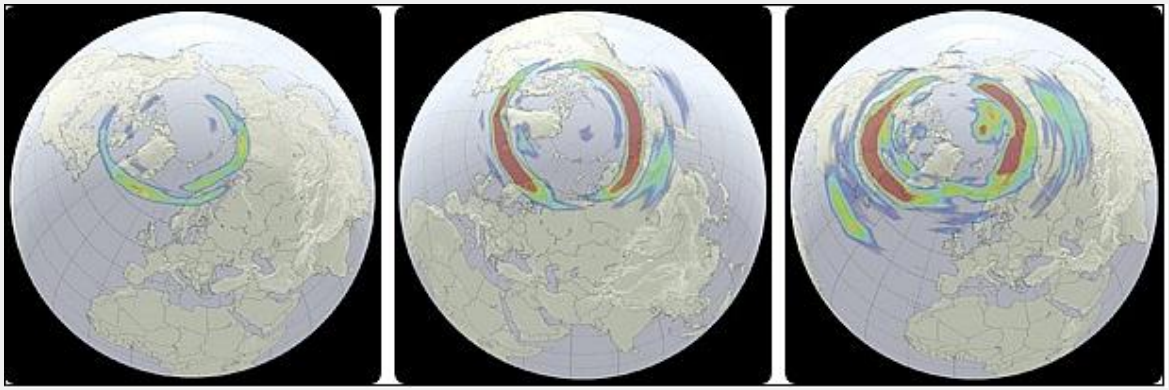


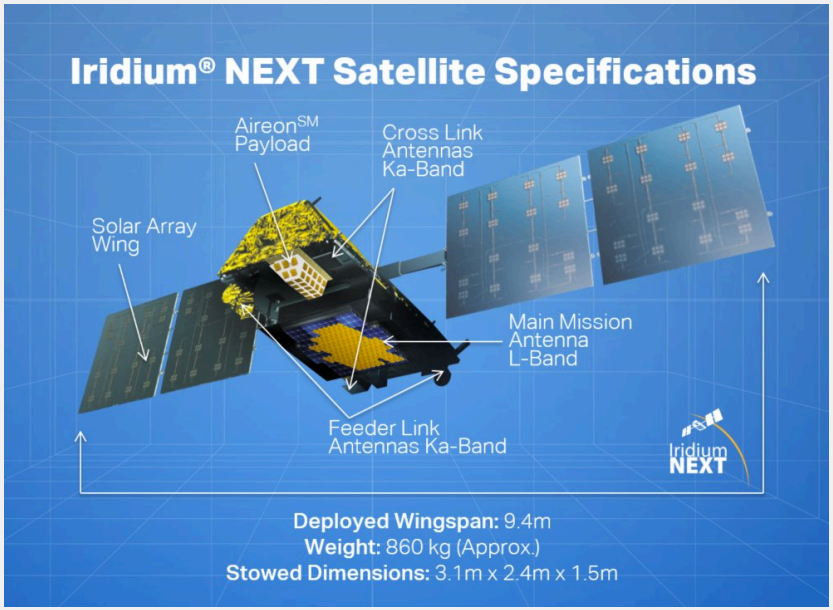


AMPERE measurements

- Iridium has a system of >80 spacecraft: 66 are active (the remaining ones function as in-orbit spares) at 780 km altitude; used for (slow) global communications
- Each Iridium satellite has a crappy magnetometer onboard
  - resolution of about 30 nT; compare to most space magnetometers with resolution << 1 nT
- But, geomagnetic storms drive field perturbations of hundreds of nT, so these can be measured!
- Now mostly replaced by **Iridium NEXT**







Hidden Figures and Re-entry

Apollo 13 Re-entry

Radio Blackout: What's going on?

- Saha Equation defines relative ionization of a gas in thermal equilibrium at temperature T:

$$\frac{n_{i+1}}{n_i} n_e = \frac{g_{i+1}}{g_i} \left(\frac{2\pi m_e kT}{h^2} \right)^{3/2} e^{-\frac{E_i}{kT}}$$

- Apollo re-entry: T ~3200 K
- Then, recall plasma frequency is related to electron density:
- Waves below plasma frequency don't get through
  - **re-entry has created a thin, temporary ionosphere around the spacecraft**

Did these movies exaggerate the effect?

- *Hidden Figures* blackout time: **1:10**
- Mercury actual blackout time:
- *Apollo 13* blackout time: **3:36** roll time; **4:30** according to dialogue
- Apollo program typical blackout time:
- Apollo 13 actual blackout time:

Exaggerated, but not too much

www.airspacemag.com/need-to-know/did-ron-howard-exaggerate-the-reentry-scene-in-the-movie-apollo-13-17639496/

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Did Ron Howard Exaggerate the Reentry Scene in *Apollo 13*?

A little bit, maybe, but not much

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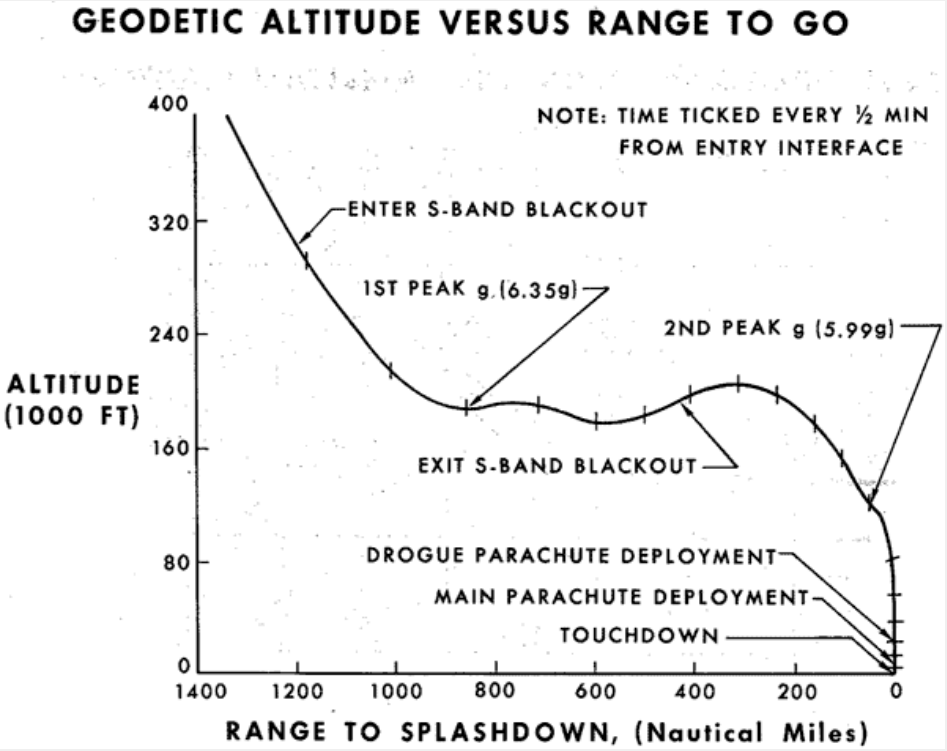
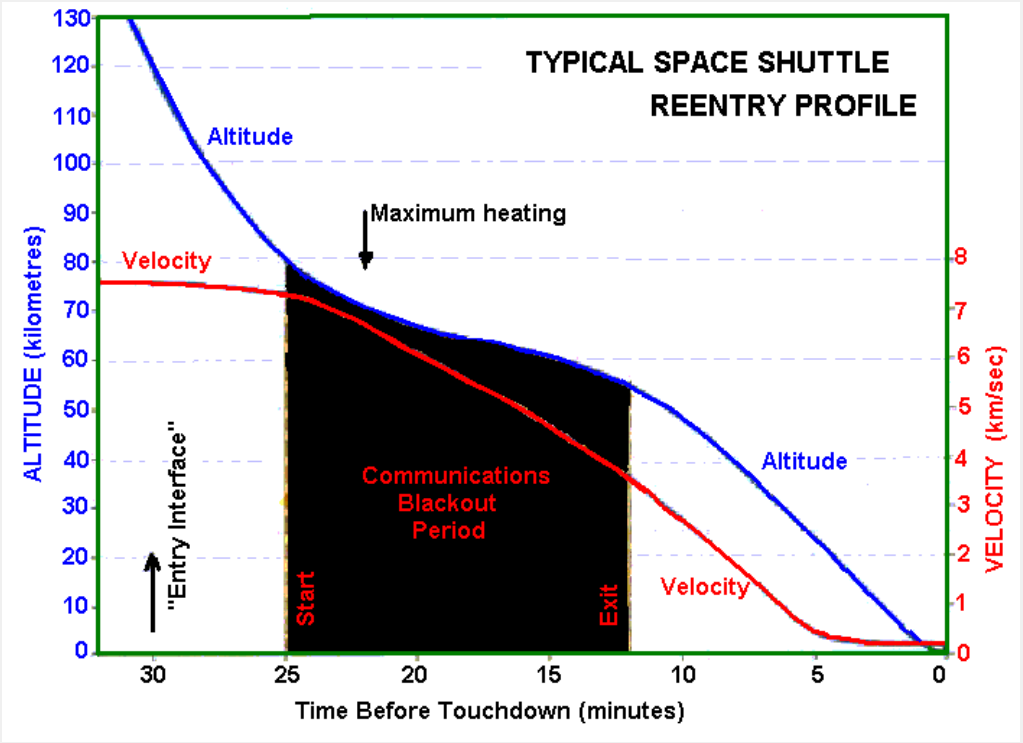
When does it occur?

- Not enough atmosphere above 100 km to create appreciable heating
- Between 50–100 km, heat gets intense enough to create significant ionization
  - Spacecraft surface gets hot, shock forms, and spacecraft rapidly decelerates
  - Once slow enough, temperature drops and ionization reduces

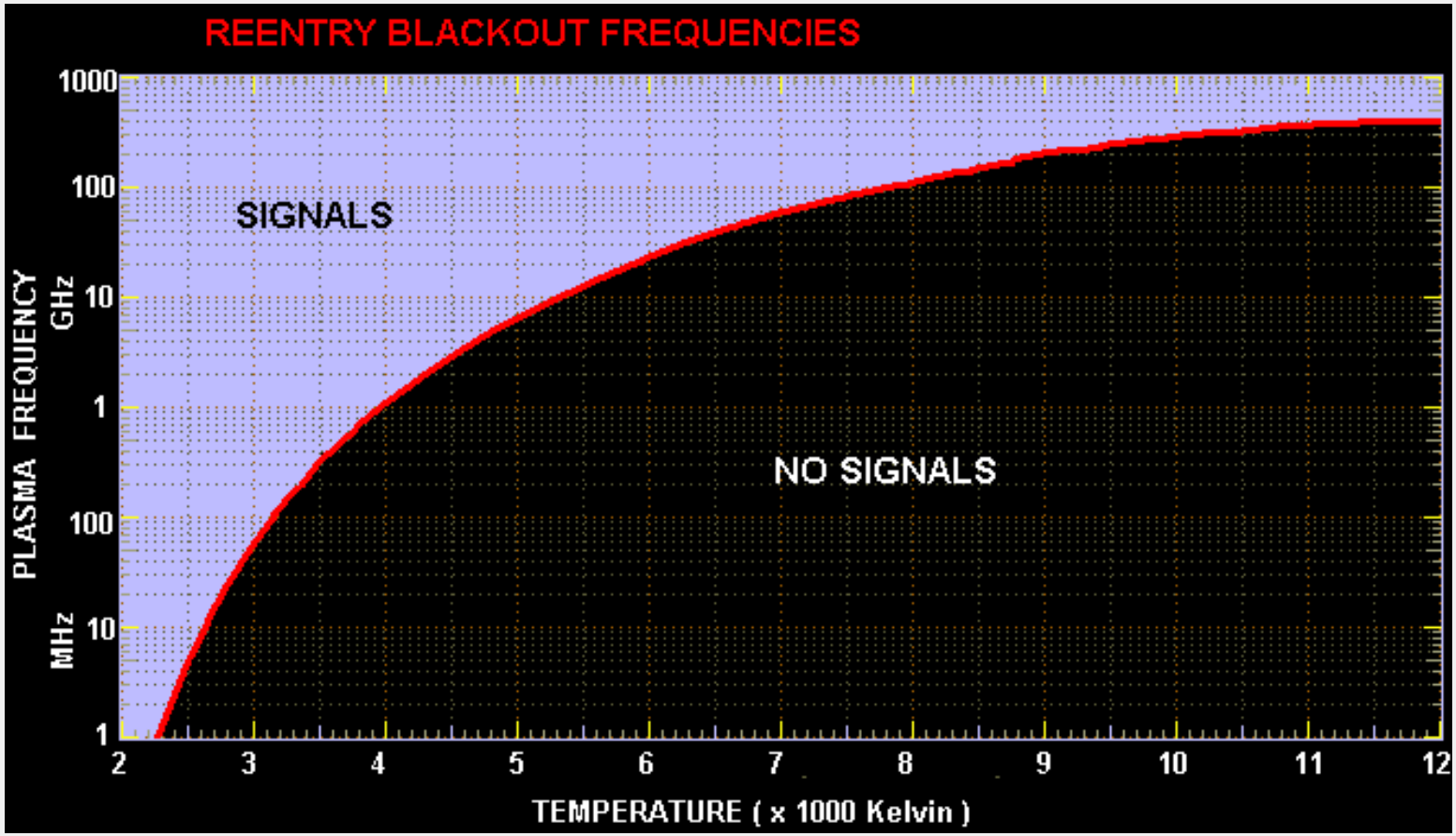




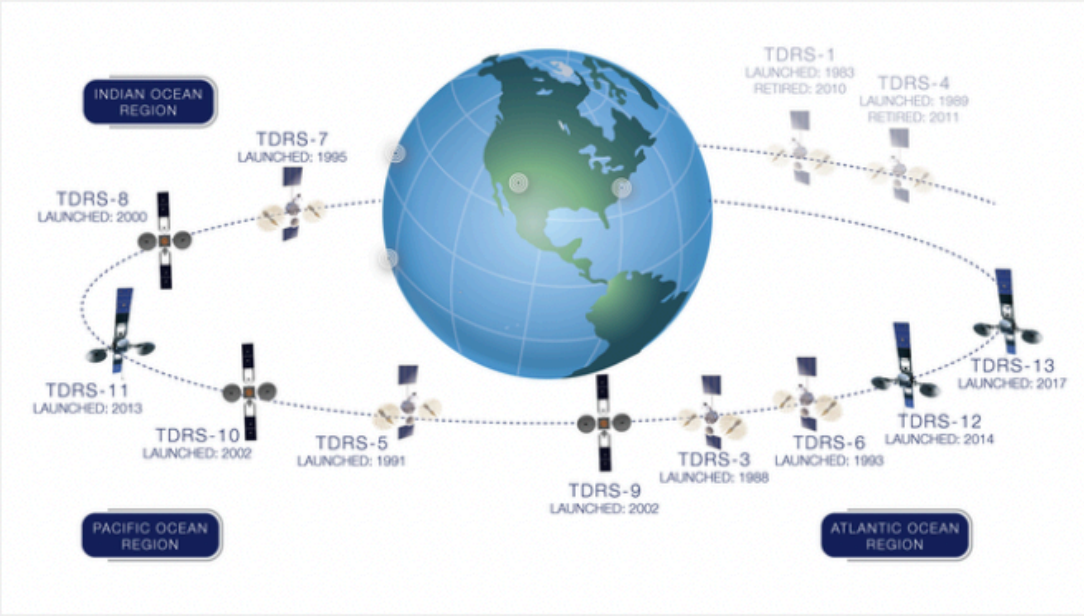
Lunar Mission Analysis Branch, Lunar Landing Branch, and Orbital Mission Analysis Branch, **"Spacecraft Operational Trajectory for Apollo Mission F, Volume I – Operational Mission Profile,"** NASA /MSC, May 17, 1969 — see Figure 5.16-3: Geodetic altitude vs. range to target (with load factor peaks, S-band blackout, parachute deployment, etc.).

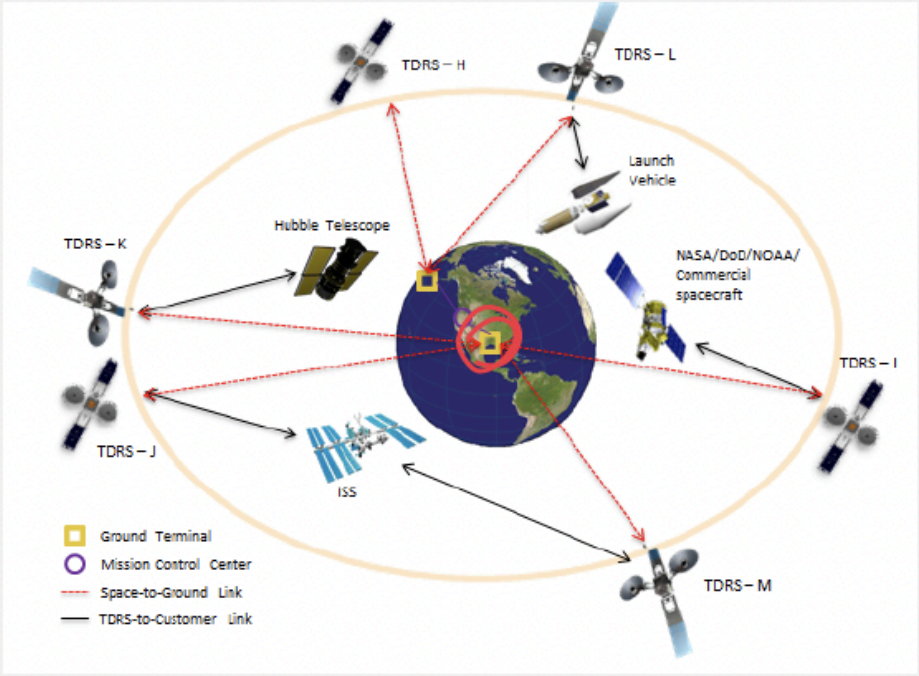
Solution 1: Use a higher frequency

- Plasma frequency is related to temperature through Saha equation
- Use a transmitting frequency higher than the plasma frequency
  - Mercury and Apollo used S-band (2.4 GHz)
  - Shuttle had a 30-minute blackout until creation in the 1980s of the Tracking and Data Relay Satellite System (TDRSS) with comms at Ku band (12-18 GHz)



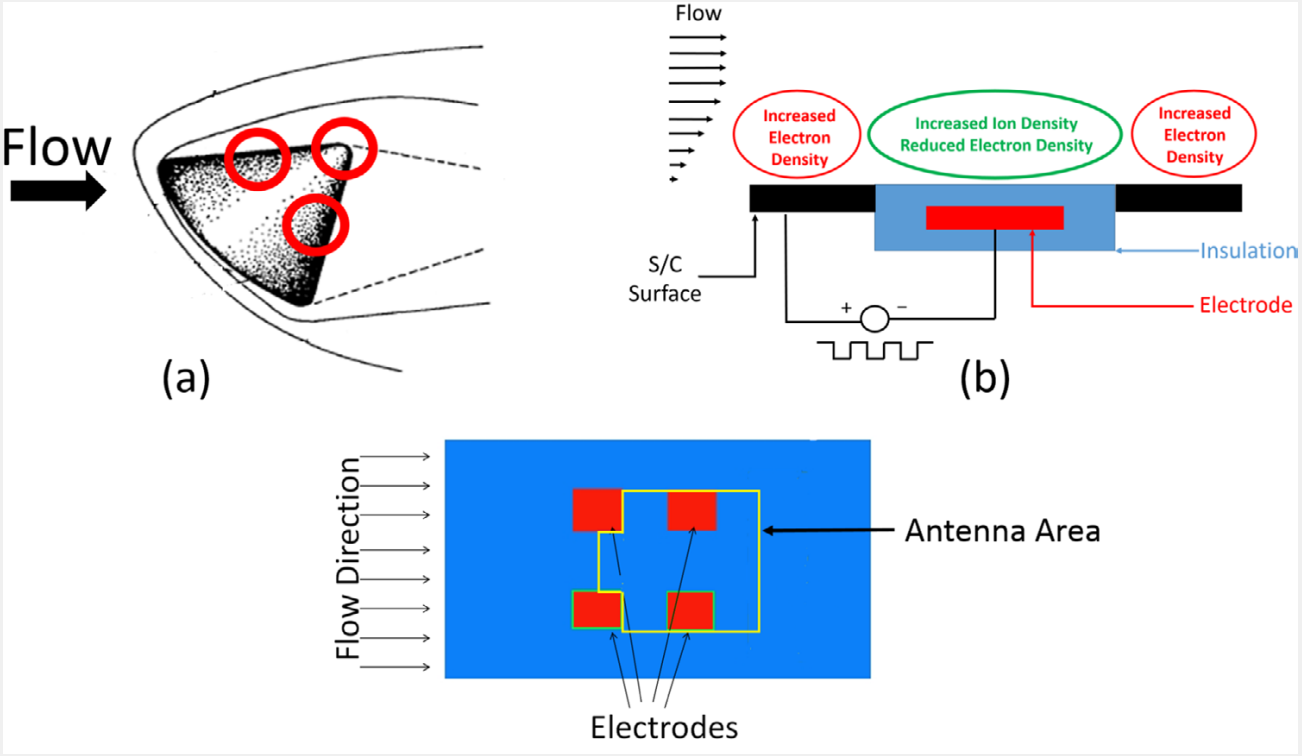
TDRSS


INDIAN OCEAN REGION
TDRS-1
LAUNCHED: 1983
RETIRED: 2010
TDRS-4
LAUNCHED: 1989
RETIRED: 2011
TDRS-7
LAUNCHED: 1995
TDRS-8
LAUNCHED: 2000
TDRS-11
LAUNCHED: 2013
TDRS-10
LAUNCHED: 2002
TDRS-5
LAUNCHED: 1991
TDRS-9
LAUNCHED: 2002
TDRS-3
LAUNCHED: 1988
TDRS-6
LAUNCHED: 1993
TDRS-12
LAUNCHED: 2014
TDRS-13
LAUNCHED: 2017
PACIFIC OCEAN REGION
ATLANTIC OCEAN REGION


TDRS – H
TDRS – L
Launch Vehicle
Hubble Telescope
TDRS – K
NASA/DoD/NOAA/ Commercial spacecraft
TDRS – I
TDRS – J
ISS
TDRS – M
Ground Terminal
Mission Control Center
Space-to-Ground Link
TDRS-to-Customer Link

Solution 2: Got any crazy ideas??

- Krishnamoorthy and Close [2017] (Stanford group)
  - Apply voltage pulses to the plasma from the spacecraft surface
  - Creates pockets where electron density is reduced 1000x



- Injecting gas or liquid, promoting recombination, thus reducing the electron density
- Use a strong magnetic field influencing the trajectory of charged particles in the plasma. This can create a "magnetic window" with lower electron density, which allows electromagnetic waves to propagate.

Up next: The Magnetosphere!

- Earth's first line of defense from the sun's harmful (charged particle) radiation.

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